Yousef Ahmed Tawfik

+965 66147686 | yousef3152003@gmail.com | Kuwait city , Kuwait | LinkedIn | GitHub | Portfolio

Professional Summary

Artificial Intelligence & Data Science graduate with hands-on experience in Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, and Generative AI. Proficient in Python, SQL, PyTorch, and TensorFlow, with a strong focus on building RAG pipelines, predictive models, and cloud-connected AI systems. Skilled at applying AI tools to debug, prototype, and solve engineering problems efficiently, with four systems already shipped to production.

Education

Arab Academy for Science, Technology and Maritime Transport (AASTMT) — Bachelor of Science in Artificial Intelligence Cairo, Egypt
2022 – 2026 | Cumulative GPA: 3.0 / 4.0 |

Publications & Research

A RAG-Enhanced Pepper Humanoid Robot for Hospital Reception and Supervised Patient Support | June 2026
Accepted at IEEE International Telecommunications Conference (ITC-Egypt 2026) — To be published in IEEE Xplore

- Co-authored a paper proposing a RAG architecture integrating the SoftBank Pepper robot with cloud AI services for patient support.
- Developed a framework bridging Pepper's legacy NAOqi platform with a modern Python 3 backend (Gemini 2.5 Flash, Whisper, FAISS).
- Documented key performance metrics: 9.97% Word Error Rate (WER) and a 2.23-second average response latency.

Selected Projects

Pepper Medical Assistant Robot — *Graduation Project & IEEE Publication* | *Sept 2025 – Present*

- Co-developed, with a 2-person engineering team, a bilingual (AR/EN) AI agent deployed and piloted with patients at a partner hospital, building the RAG pipeline (FAISS + Gemini 2.5 Flash) and Whisper-based speech recognition for hospital reception and patient triage.
- **KPIs**: 2.23s average response time, 100% API success across supervised hospital test sessions, 9.97% bilingual Whisper WER, RAG grounded over 70 doctors across 16 departments and 8 branches.
- **Technologies**: Python, Flask, FAISS, Whisper, Gemini 2.5 Flash, RAG, Robotics.

Market Intelligence Dashboard — *Full-Stack Retail Analytics Platform (Production)*

- Built a production retail analytics dashboard processing 84,000+ rows across 14 categories and 17 regional markets, with a FastAPI backend (10+ endpoints) and an adaptive segmentation algorithm that scales to new categories with zero code changes.
- **Technologies**: Python, FastAPI, Pandas, JavaScript, Chart.js.

Smart Sales Assistant — *Bilingual AI Sales Agent (Production)*

- Built a production bilingual (Arabic/English) LLM sales agent using function-calling with 12 tools for retrieval, cart, and order management over a 1,145-product catalogue, with grounded retrieval (FAISS) eliminating hallucinated products and prices.
- **Technologies**: LLM Agents, Function Calling / Tool Use, RAG, FAISS, FastAPI, Docker.

Audio Intelligence Platform — *Arabic Call Analytics Platform (Production)*

- Built a hybrid cloud/local Arabic call-analysis platform with automatic speech-to-text failover, speaker diarization, MARBERT sentiment analysis, and LLM-generated manager summaries, backed by a 131-test CI suite with automated secret scanning.
- **Technologies**: FastAPI, faster-whisper, pyannote, Transformers, Docker, CI/CD.

Technical Skills

Programming Languages: Python, C, SQL, Dart (Flutter), HTML, CSS
AI & Data Science: Machine Learning (ML), Deep Learning (DL), NLP, Computer Vision (CV), LLMs, Data Mining, Big Data Analytics
Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, YOLO, NumPy, Pandas, OpenCV, FastAPI
Advanced AI & Generative AI: Generative AI, Retrieval-Augmented Generation (RAG), FAISS, LLM Agents, Function Calling / Tool Use, Google Gemini, Whisper ASR, Prompt Engineering
Mobile Development: Flutter, Firebase (Auth, Realtime Database), Provider, GoRouter
Tools & Technologies: Git, GitHub, Android Studio, Jupyter, Google Colab, SQLite, REST APIs, Cloud Computing, HPC, Docker, Web Scraping, Chart.js

Certifications

- **Akhnaton Trading & Distribution S.A.E:** Summer Internship Program, Digital Transformation Department | *Jul 1, 2026 – Jul 30, 2026*
- **AMIT Learning:** Master Machine Learning & AI Diploma (220 Hours, Grade: 92%) | *Feb 18, 2025 – Sep 25, 2025*
- **Information Technology Institute (ITI):** Machine Learning Training Program (60 Hours) | *Aug 14, 2025 – Aug 29, 2025*
- **The British University in Egypt (BUE):** Applications of Artificial Intelligence (15 Hours) | *Sep 6, 2025 – Sep 8, 2025*